

Amogh Rahul Rathod

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Experience

Backend / AI Systems Developer

Jun 2026 - Present \ Nexaneura Technologies Pvt Ltd \

- Built and deployed a self-hosted AI voice agent for scheduled outbound calls via Twilio, self-hosted on GCP; achieved <1.6s latency on international (US) test calls.
- Added multilingual conversational support for Indian languages including Hindi, Gujarati, and Marathi.
- Deployed and managed the system on Google Cloud Platform, including scaling and reliability.

Backend Developer (Founding Team)

Nov 2024 - Mar 2025 \ Guide, Bengaluru \

- Built the backend from scratch, defining API contracts and core authentication flows.
- Developed a Go authentication server with REST APIs, JWT authentication, salted password hashing, and PostgreSQL.
- Structured the backend into modular services for integration with the React frontend.

Education

B.Tech Computer Science (AIML)

Aug 2023 - Aug 2027 (Expected) \ PES University, Bengaluru \ CGPA: 8.2/10 \

- Prof. CNR Rao Scholarship (Top 20%) (2x)
- Distinction Award (3x)

Skills & Abilities

Languages: C/C++, Java, Python, Go, Rust

Cloud, DevOps & Systems: Docker, Kubernetes, Container Internals (Namespaces, chroot), AWS (EC2, VPC, IAM), GCP, Git

ML / MLOps: Reinforcement Learning (PPO), Gymnasium, Seldon Core, Kestra, CodeCarbon \ **Security & Networking:** Penetration Testing, Wireshark, Tshark, Aircrack-ng suite, Wireless Protocol Analysis, Flipper Zero \ **Backend & MVC:** Spring Boot, Spring MVC, REST API Design, Node.js, Express, PostgreSQL, ORM (JPA/Hibernate)

Projects

Docksmith – Container Build & Runtime Engine (Python)

2026 \ [repo-link](#)

- Built a Docker-like image builder and container runtime from scratch in Python using content-addressed layers, namespaces, and chroot.
- Added deterministic build caching using layer, instruction, and source-file hashes.
- Tested container isolation using separate mount/PID namespaces.

GreenOps – RL-Driven Sustainable MLOps Decision System

2026 \ private-repo

- Built a PPO-based RL agent that chooses between retraining, fine-tuning, and skipping training based on model drift and performance.
- Developed a Gymnasium environment using PSI, cosine distance, model accuracy, and CodeCarbon energy data.
- Integrated the agent with Seldon Core and Kestra to automate model maintenance decisions.

KawaiiFlix – Netflix-Style Streaming Platform (Java Microservices)

2026 \ [Backend](#) | [Frontend](#) \

- Built a Netflix-style streaming platform with Java 21, Spring Boot, React, and a microservices architecture.
- Worked on an API Gateway with separate auth and content services backed by PostgreSQL.
- Implemented JWT authentication, MinIO video storage, and Docker Compose deployment.

GTKWave Open Source Contribution (macOS Build Fix)

2024 [Merged PR](#) \

- Fixed macOS build issues in GTKWave and updated the upstream documentation; the changes were reviewed and merged into the official repository.

Achievements

- Ranked 31st out of 300+ teams in NITK Code-Red Cybersecurity CTF.
- Arduino Robotics Certifications – Foundation to Master level.